MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, ENGINE EXHAUST, TYPE MJ-4

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of single pointer engine exhaust thermocouple, temperature indicator, designated Type MJ-4.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.3), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to extent specified herein.

SPECIFICATIONS

FEDERAL

* PPP-B-1672	Box, Shipping, Reusable, with Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-I-7071	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection by Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-147	Palletized and Containerized, Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS35338	Washer-Lock Spring, Helical, Regular (Medium) Series
MS33639	Case, Instrument, Clamp-Mounted, Aircraft.

Beneficial comments (see recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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AIR FORCE-NAVY AERONAUTICAL

AN961 Washer-Flat, Electrical

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 Preproduction. The indicator furnished under this specification shall be a product which has been tested and has passed the tests specified in MIL-I-7071B or as specified herein.

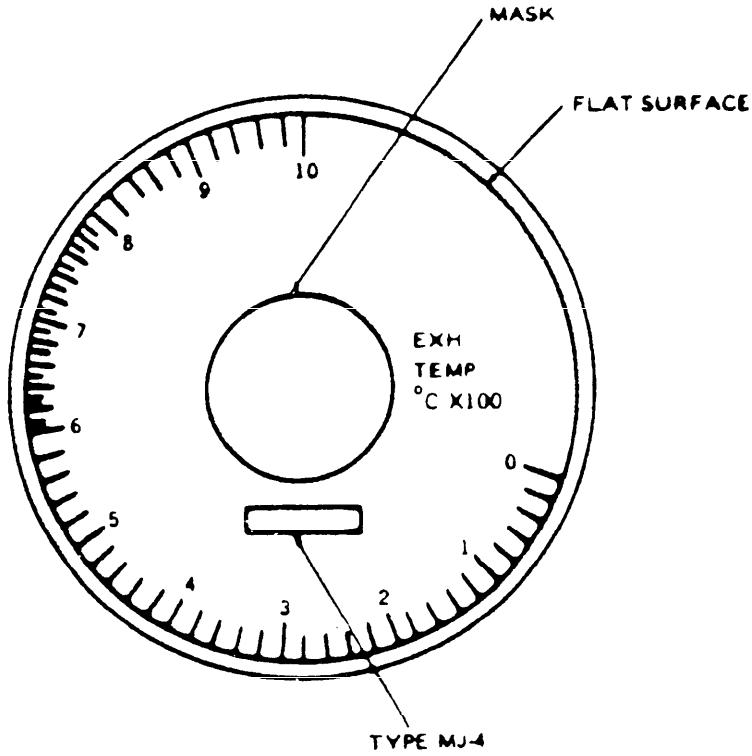
3.2 General specification. The requirements of MIL-I-7071 apply as requirements of this specification. Additional requirements shall be as specified herein.

3.3 Design and construction.

3.3.1 The indicator shall conform to Figures 1 and 2.

3.3.2 Case back. The back of the indicator case shall be distinctly marked to indicate "+" for the chromel terminal stud and "-" for the alumel terminal stud. The statement "USE 8 OHM C-A LEADS" shall be plainly marked on the back of the case.

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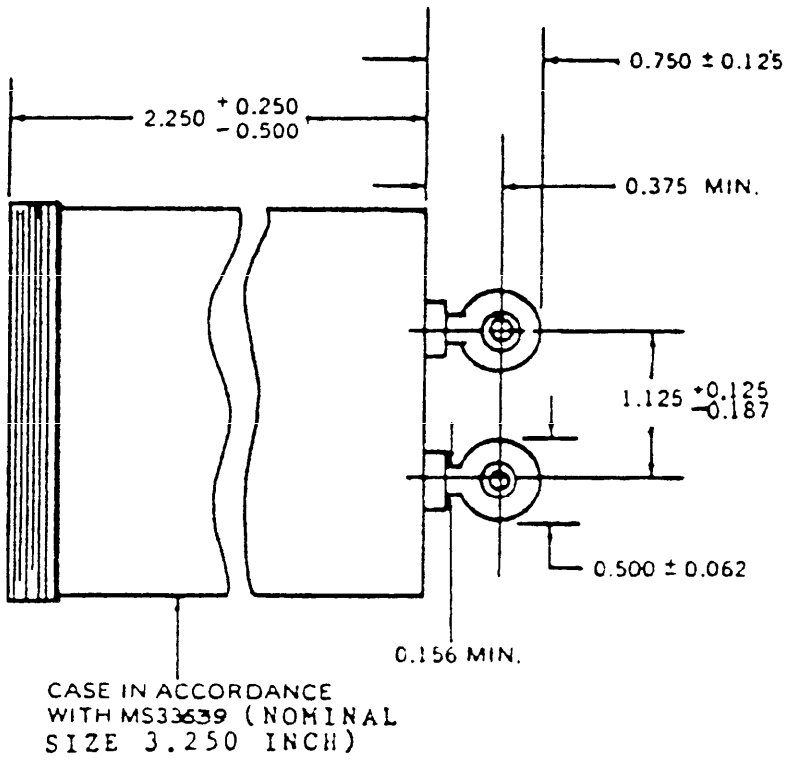


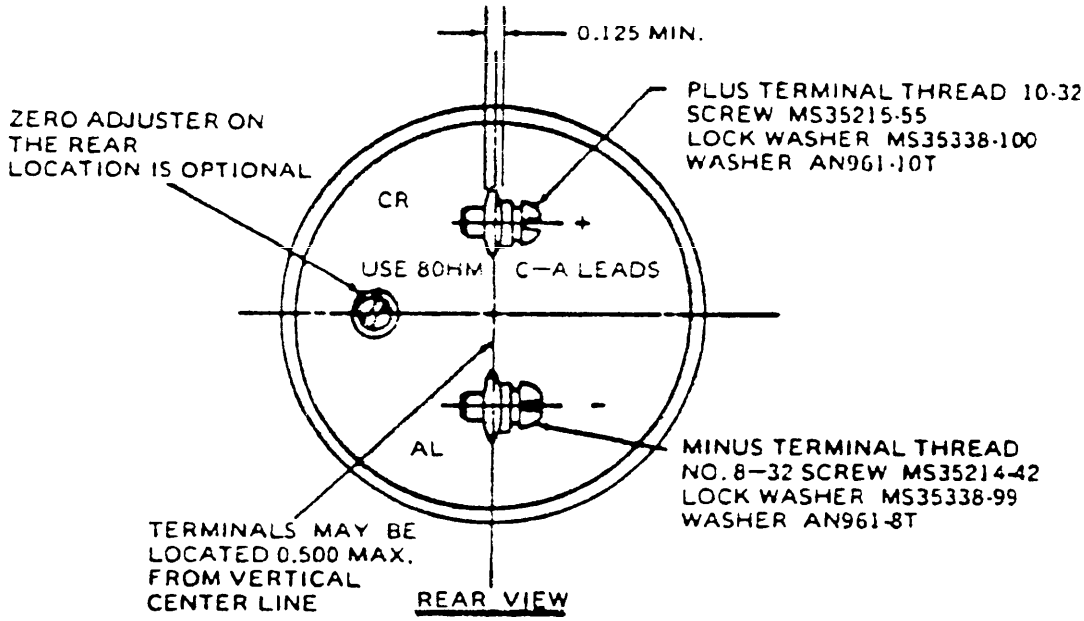
MARKING	HEIGHT OR LENGTH ±0.005	WIDTH OF LINE OR GRADUATION ± 0.005	MATERIAL OR FINISH
NUMERALS	0.156	0.020	FLUORESCENT LUMINESCENT MATERIAL
100° C GRADUATIONS	0.219	0.020	
20° C GRADUATIONS	0.156	0.010	
10° C GRADUATIONS	0.094	0.010	
LETTERING			
"EXH TEMP"	0.156	0.020	
"°C X 100"	0.125	0.010	
VISIBLE PORTION OF POINTER	———	———	
TYPE NO.	0.062	0.010	DURABLE DULL BLACK
BACKGROUND OF DIAL	———	———	

DIMENSIONS IN INCHES

FIGURE 1. DIAL

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BREAK SHARP EDGES APPROX. 0.010 RADIUS
DIMENSIONS IN INCHES.

FIGURE 2. CASE

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3.3.3 Dial. The dial of the indicator shall conform to Figure 1.

3.3.3.1 Scale arc. The scale arc of the indicator shall not be less than 250 degrees. The length of the scale arc shall not be less than 6.25 inches as measured on the outer arc of the scale. The indicator mechanism shall not be geared to obtain the required scale arc.

3.3.3.2 Scale distribution. The scale of the indicator shall be as linear as is practical and shall not be contracted by more than 20 percent between any two adjacent scale graduations.

3.3.4 Pointer. The width of the pointer 1/2 ± 1/16 inch from the tip shall be 5/64 ± 1/64 inch and shall taper to a point.

3.3.4.1 Pointer overlap. The pointer length shall be such that the pointer tip will extend into the longest scale graduation a distance of 3/32 ± 1/64 inch.

3.3.4.2 Pointer overswing. The pointer overswing shall not be less than 3 angular degrees beyond each end of the scale.

3.4 Weight. The weight of the complete indicator shall not exceed 1.20 pounds.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test conditions. The tolerances specified in Table I, not enclosed in parenthesis, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.2.1 General. Inspection, and testing of the indicator shall be in accordance with MIL-I-7071, and as specified herein.

4.2.2 Sampling. Lot formation, sampling, acceptance or rejection, and retest shall be in accordance with MIL-STD-105.

4.2.2.1 Sampling Plan A shall be Inspection Level II, AQL 2.5 percent.

4.2.2.2 Sampling Plan B shall be Inspection Level S-1, AQL 6.5 percent.

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TABLE I. TEST TOLERANCES

TEST POINT	SCALE ERROR AT ROOM TEMPERATURE	SCALE ERROR AT LOW TEMPERATURE	SCALE ERROR AT HIGH TEMPERATURE	SCALE ERROR AT ROOM TEMPERATURE AFTER VIBRATION FAILURE	EXTREME LOW TEMPERATURE EXPOSURE	
					SCALE ERROR AT LOW TEMPERATURE	SCALE ERROR AT ROOM TEMPERATURE AFTER LOW TEMPERATURE EXPOSURE
°C	TOLERANCE °C	TOLERANCE °C	TOLERANCE °C	TOLERANCE °C	TOLERANCE °C	TOLERANCE °C
0	60 (60)	75 (75)	70 (70)	75 (70)	90 (75)	80 (65)
200	40 (40)	55 (55)	50 (50)	55 (50)	70 (55)	60 (45)
400	30 (30)	45 (45)	40 (40)	45 (40)	60 (45)	50 (35)
500	20 (20)	35 (35)	30 (30)	35 (30)	50 (35)	40 (25)
600	10 (10)	25 (25)	20 (20)	25 (20)	40 (25)	30 (15)
700	5 (5)	20 (20)	15 (15)	20 (15)	35 (20)	25 (10)
800	10 (10)	25 (25)	20 (20)	25 (20)	40 (25)	30 (15)
900	20 (20)	35 (35)	30 (30)	35 (30)	50 (35)	40 (25)
1000	20 (20)	35 (35)	30 (30)	35 (30)	50 (35)	40 (25)

TEST POINT	SCALE ERROR AT ROOM TEMPERATURE AFTER HIGH TEMPERATURE EXPOSURE	SCALE ERROR AT ROOM TEMPERATURE AFTER SALT SPRAY
°C	TOLERANCE °C	TOLERANCE °C
0	80 (65)	80 (60)
200	60 (45)	60 (40)
400	50 (35)	50 (30)
500	40 (25)	40 (20)
600	30 (15)	30 (10)
700	25 (10)	25 (5)
800	30 (15)	30 (10)
900	40 (25)	40 (20)
1000	40 (25)	40 (20)

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* 4.3 Preproduction tests. The preproduction tests shall consist of the individual tests and those specified under sampling Plan A and sampling Plan B.

4.3.1 Individual tests.

4.3.1.1 Scale error at room temperature. The indicator pointer shall first be set to read 700° Centigrade (C) when the millivoltage for 700°C, as specified in Table I of MIL-I-7071, for the ambient room temperature is applied to the terminals of the indicator through the specified resistance in series with the indicator. The test shall then be conducted at the specified test points of Table I herein. The scale errors shall not exceed the tolerances specified. If greater overall indicator accuracy can be obtained by setting the pointer to 700 ± 5°C with the specified voltage for 700°C, the pointer shall be so positioned.

4.3.1.2 Position error. This test may be conducted at any point on the scale between 400° and 900°C by applying test voltage, the maximum change in indication shall not exceed 25°C.

4.3.1.3 Friction. The maximum difference in reading, before and after tapping shall not exceed 30°C.

4.3.1.4 Zero adjustment. Power shall be applied to produce a 700°C reading. When the indicator zero adjustment is operated throughout its range, the resulting pointer motion shall be not less than 40°C above and below the 700°C reading. The total motion of the pointer shall not exceed 200°C. The pointer shall be reset to the initial reading with the initial power applied.

4.3.1.5 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

4.3.2 Sampling plan A.

* 4.3.2.1 Scale error at low temperature. The scale errors shall not exceed the tolerances specified in Table I when subjected to the test requirements of paragraph 4.6.8 of MIL-T-7071. If there is any tendency for the pointer to stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

* 4.3.2.2 Scale error at high temperature. The scale errors shall not exceed the tolerances specified in Table I when subjected to the test requirements of paragraph 4.6.9 of MIL-I-7071. If there is any tendency for the pointer to stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

* 4.3.2.3 Magnetic effect. When the indicator is subjected to the magnetic effect test, as specified in paragraph 4.6.10 of MIL-I-7071, it shall not cause the magnetic compass to deflect more than 5 degrees.

* 4.3.2.4 Vibration error. The indicators shall be subjected to the test requirements of paragraph 4.6.11 of MIL-I-7071. The maximum amplitude of pointer oscillation throughout the frequency range shall not exceed a range of 20°C within the temperature range of 400° to 900°C and shall not exceed 30°C for the remainder of the scale. The maximum difference between the original position of the pointer before vibration and the position of the pointer during

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vibration shall not exceed 15°C between 400° and 900°C and shall not exceed 25°C for the remainder of the scale. Pointer oscillation shall not be considered as pointer variation. The diameter of the circular motion of vibration shall be 0.009 to 0.011 inch.

* 4.3.2.5 Vibration failure. The scale errors at room temperature after this test, as specified in paragraph 4.6.12 of MIL-I-7071, shall not exceed the tolerances specified in Table I. The zero adjustment shall not be reset after this vibration test.

4.3.2.6 Resistance. The resistance of the indicator shall be not less than 40 ohms.

4.4 Sampling plan B. The tests covered by the sampling plan B shall be the following:

4.4.1 Cold junction compensator lag. The indicator shall be connected to a chromel-alumel thermocouple (8-ohms total external circuit resistance), mersed in an ice bath of 0°C, and a millivoltage of constant value applied in series with the indicator and thermocouple to provide readings of 700°C. The indicator shall then be subjected to the conditions specified in MIL-I-7071 for increasing and decreasing temperatures. The indication of the indicator throughout the test shall at no time vary by more than 30°C from the indication obtained at room temperature.

4.4.2 Extreme low temperature exposure. The scale errors at low temperature, tests as specified in paragraph 4.6.15 of MIL-I-7071, shall not exceed the tolerances specified in Table I. The scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table I.

4.4.3 High temperature exposure. The scale errors at room temperature after high temperature exposure tests, as specified in paragraph 4.6.16 of MIL-I-7071, shall not exceed the tolerances specified in Table I.

4.4.4. Thermal shock. No damage to the indicator shall occur as result of this test when subjected to the test requirements of 4.6.18 of MIL-I-7071. Internal fogging of the indicator glass shall be sufficient cause for rejection.

4.4.5 Salt spray. The scale errors at room temperature after salt spray exposure tests, as specified in paragraph 4.6.20 of MIL-I-7071, shall not exceed the tolerances specified in Table I.

4.5 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C or industrial as specified (see 6.3).

5.1.1 Cleaning. Indicators shall be cleaned in accordance with MIL-P-116, process C-1.

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5.1.1.2 Drying. Indicators shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation applications. Preservatives shall not be used.

5.1.1.4 Unit Packs. Each indicator shall be individually packaged in accordance with MIL-P-116, Method III. Unit packs shall be individually packed in containers conforming to PPP-B-1672.

5.1.2 Level C. The level C preservation for indicators shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Industrial. The industrial preservation of indicators shall be in accordance with the requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B, C, or industrial, as specified (see 6.3).

5.2.1 Level A. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level A protection.

5.2.2 Level B. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level B protection.

5.2.3 Level C. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level C protection.

5.2.3.1 Industrial. The packaged indicators shall be packed in accordance with the requirements of ASTM-D-3951.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent possible. Unitized loads shall be in comformance with MIL-STD-147.

5.3 Marking.

5.3.1 Level A, B, C. In addition to any special or other identification marking required by the contract (see 6.3), each unit pack exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial. Industrial marking shall be accordance with the requirements of ASTM-D-3951.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible.

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6. NOTES

6.1 Intended use. The thermocouple temperature indicator covered by this specification is intended to indicate the exhaust temperature of jet engines.

6.2 This indicator can be used on instrument panels requiring case mounting in accordance with MS27572 by application of an instrument mounting clamp conforming to MIL-C-6359.

6.3 Ordering data.

6.3.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Levels of packaging and packing required (see section 5).

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 82

Review Activity
DLA-GS

Project No. 6685-F732

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-25026B(USAF)	Indicator, Temperature, Thermocouple, Engine Exhaust, Type MJ-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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